

SourcePilot AI

System Architecture, Tech Stack & Feature Specification Overview

System Overview: SourcePilot AI is an autonomous, agentic B2B procurement platform designed to streamline the complete enterprise sourcing lifecycle—from natural language requirement extraction and 5-factor supplier ranking to multi-approver RFQ governance, user-scoped OAuth email automation, and Quote Management.

1. Technology Stack Architecture

SourcePilot AI utilizes a modern, decoupled microservice architecture structured for security, scalability, and deterministic AI execution.

Layer	Technology / Framework	Key Purpose & Implementation Details
Frontend UI	React 18, TypeScript, Vite, Lucide Icons	Modern reactive SPA with Glassmorphism dark UI theme, dynamic tab navigation, and live status badges.
Backend API	Python 3.11, FastAPI, Pydantic v2	Asynchronous REST API framework with strict request/response data validation and OpenAPI schema generation.
Database & ORM	SQLite / PostgreSQL, SQLAlchemy ORM	Relational database layer with strict foreign key constraints, automated model creation, and schema migrations.
AI Orchestration	Google Gemini 2.5 Flash API	Multi-agent AI suite for specification extraction, Phase 2 supplier ranking, email drafting, quote parsing, and recommendations.
Connected Mailboxes	Gmail API v1, Outlook Graph API	User-scoped OAuth 2.0 PKCE authentication for sending approved RFQs directly from personal connected mailboxes.
Security & Crypto	Fernet 32-byte Key, JWT, OAuth PKCE	Symmetric Fernet encryption for OAuth access/refresh tokens stored at rest; state token validation against CSRF.
SMTP Fallback	Mailtrap Sandbox SMTP	Automated fallback email dispatch engine when no personal mailbox is connected.

2. End-to-End System Workflow & Capabilities

Phase 1 — Requirement Extraction & Specification Parsing

Buyers input unstructured sourcing requests (e.g. *'500 Industrial Brushless DC Motors for automotive assembly'*). The Requirement Extraction Agent parses these into structured specifications including target quantities, unit prices, technical tolerances, and compliance standards.

Phase 2 — Autonomous Supplier Discovery & Phase 2 Multi-Factor Ranking

SourcePilot queries stored supplier databases and calculates deterministic Phase 2 scores across 5 weighted factors:

- **Spec Match (35%):** Technical parameter alignment against extracted specifications.
- **Certification (25%):** ISO, CE, RoHS, and industry compliance verification.
- **Geographic Alignment (15%):** Regional logistics proximity and shipping feasibility.
- **Contact Feasibility (15%):** Valid email and phone contact readiness.

- **Source/Data Quality (10%)**: Completeness and freshness of supplier profile data.

Re-ranking endpoint (POST /suppliers/rerank) allows re-evaluating existing suppliers without triggering external search APIs.

Phase 3 — RFQ Generation, Revision Control & Multi-Approver Governance

- **RFQ Generation**: Generates professional Markdown RFQ documents with sanitized titles and organization metadata.
- **Revision Control**: PATCH /rfq/{rfq_id} automatically increments document revision counters (v1, v2, v3).
- **Multi-Approver Workflow**: Configurable role-based approval steps (Buyer → Approver → Director) requiring explicit sign-off before dispatch.

Phase 4 — User-Scoped Email Automation & OAuth Integration

- **Email Drafting**: Converts Markdown RFQs into clean plain-text email bodies, dynamically populating recipient supplier emails.
- **Subject Line Sanitization**: Cleans duplicate requester prefixes into standard titles (e.g. [RFQ] Invitation to Quote: 500.0x Motors).
- **OAuth Mailbox Integration**: Connects personal Gmail accounts via Google OAuth 2.0 PKCE, encrypting tokens securely with Fernet.

Phase 5 — Quote Management & Award Lifecycle

- **Quote Capture (Phase 1)**: Manual modal (AddQuoteModal.tsx) and PDF/XLSX file parser for recording quotes with validation (prices > \$0, lead time ≥ 0).
- **Side-by-Side Comparison (Phase 2)**: Grid rendering Quote Ref, Status, Unit Price, Total Quote, Lead Time, MOQ, Terms, Warranty, and Expiry.
- **Human-in-the-Loop Award**: POST /quotations/award/{requirement_id} sets awarded quote status to accepted and non-awarded quotes to rejected.

3. Database Schema & Data Models

Table Name	Key Fields & Types	Relationships & Purpose
procurement_requirements	id, title, raw_text, structured_data (JSON), status, organization_id	Root procurement request record. Links to RFQs, matches, quotations, and recommendations.
suppliers	id, company_name, location_country, website, trust_score, verified	Global supplier registry. Links to matches, contacts, dispatches, and quotations.
requirement_supplier_matches	requirement_id, supplier_id, match_score, ranking_explanation (JSON)	Junction table storing Phase 2 5-factor scoring results per supplier match.
rfqs	id, requirement_id, title, content, version, status, approved_at	RFQ document record. Versioned upon updates; links to approvals and dispatches.
rfq_approvals	id, rfq_id, step_number, role_required, status, approver_id, decided_at	Multi-approver governance chain tracking approval decisions.
connected_accounts	id, user_id, provider, email_address, encrypted_access_token, status	User-scoped OAuth connected mailboxes (Fernet encrypted tokens).
quotations	id, rfq_id, requirement_id, supplier_id, quote_reference, status, extracted_data	Supplier quotation records storing prices, lead times, warranty, and status ('accepted'/'rejected').
recommendations	id, requirement_id, summary, recommended_supplier_id, awarded_supplier_id	AI Executive Summary & human award sign-off record.